Spotify Streaming Behavior Analysis

1. Project Overview

This project analyzes personal Spotify streaming history to uncover listening behaviors, music preferences, artist discovery patterns, and long-term engagement trends. Using raw Spotify Extended Streaming History, the project demonstrates an end-to-end analytics workflow involving data cleaning, feature engineering, API enrichment, SQL-based analysis, and interactive dashboard development.

The primary objective was to transform raw JSON streaming logs into a structured analytical dataset capable of answering meaningful behavioral questions through visualization and data-driven insights.

The end-to-end analytics workflow includes:

- **Python:** Data cleaning, preprocessing, feature engineering, and JSON processing.
- **Spotify Web API:** Retrieval of artist genres, popularity scores, and album artwork to enrich the streaming dataset.
- **SQL Server:** Structured data storage and analytical querying to identify listening trends and behavioral patterns.
- **Power BI:** Development of an interactive Spotify-inspired dashboard to communicate insights through data visualization.

2. Dataset Summary

- Data Source: Personal Spotify Extended Streaming History (JSON)
- Initial Structure:
    - Rows: 39,568 raw multi-year streaming events
    - Columns: 23 metadata fields per event

    The raw dataset contained a mix of track-level, episode-level, and audiobook-related metadata, along with several sparsely populated fields.

- **Platform Standardization:** Highly granular platform values (OS versions, device models) were standardized into a new **device_type** column (Mobile, PC/Desktop, Smart Speaker, Others) to enable clean, high-level analysis while preserving original device data for granularity.

3. Exploratory Data Analysis (EDA) & Data Cleaning

Before analysis, an exploratory data analysis (EDA) was performed to understand data quality, identify inconsistencies, and prepare the dataset for analytical processing.

➢ **Initial Data Quality Assessment**

A null-value analysis was conducted to evaluate data completeness:

- A total of **57 records** contained missing values across essential attributes such as Track, Artist, Album, and Spotify Track URI. Since these fields uniquely identify each streaming event, the incomplete records were removed to maintain analytical integrity.
- Several non-music-related columns were found to be almost entirely null, including:
  - Episode Name
  - Episode Show Name
  - Spotify Episode URI
  - Audiobook Title
  - Audiobook URI
  - Audiobook Chapter URI
  - Audiobook Chapter Title

  Since the analysis focuses on music listening behavior, these columns were removed to reduce noise and improve dataset relevance.

➢ **Column Standardization & Renaming**

To improve readability and usability:

- Renamed verbose metadata fields to concise, analysis-friendly names:
  - master_metadata_track_name -> track
  - master_metadata_album_artist_name -> artist
  - master_metadata_album_album_name -> album
  - reason_start -> start_reason
  - reason_end -> end_reason

- Converted the original timestamp column to DATETIME format and renamed it to:
  - event_time

- **Feature Engineering**

Several analytical features were engineered to support behavioral analysis:

- **Year, Month, Day of Week, Hour** – enabled temporal trend analysis.
- **Listening Minutes** – converted raw milliseconds into a more interpretable measure.
- **Day Part** – grouped listening into Morning, Afternoon, Evening, and Night.
- **Is Weekend** – supported weekday versus weekend comparison.
- **Season** – enabled seasonal listening analysis.

The original platform information was preserved while enabling **clean, high-level device analysis**.

4. API Enrichment Using Python

To enrich the streaming history with additional contextual information, the Spotify Web API was integrated into the workflow. Artist metadata including genres, popularity scores, and album artwork was retrieved and merged with the cleaned dataset.
This enrichment enabled analysis beyond listening duration by incorporating artist characteristics and improving dashboard storytelling.

5. Data Analysis Using SQL Server

Following preprocessing, the refined dataset was imported into **SQL Server** where analytical queries were developed to investigate listening behavior, artist preferences, engagement patterns, and device usage.
The SQL analysis focused on answering business-style analytical questions rather than simply summarizing the data

The analysis focused on four major themes:

- Overall listening patterns
- Artist and music preference analysis
- Time-based listening behavior
- Device usage and engagement patterns

Below is a summary of each insight derived from the SQL queries.

1) Total Listening Time

Listening analysis shows a high cumulative number of listening hours, indicating that Spotify is a consistently used platform rather than an occasional entertainment source.

total_listening_hours
1979

2) Year-over-Year Listening Trends

Yearly listening patterns remain relatively stable over time, suggesting habitual usage rather than sharp growth or decline driven by short-term trends.

	year	listening_hours
1	2022	302.46
2	2023	539.62
3	2024	553.73
4	2025	583.79

3) Average Daily Listening Time

Average daily listening duration reflects regular engagement with music, supporting the idea that Spotify is integrated into everyday routines.

avg_daily_minutes
105.49

4) Top Artists Driving Listening Time

A small group of top artists accounts for a disproportionately high share of total listening hours, highlighting strong preference concentration among favorite artists.

	artist	listening_hours
1	SEVENTEEN	80.76
2	BTS	74.32
3	Taylor Swift	50.49
4	Gibran Alco...	41.65
5	Shin Giwon ...	34.44
6	A.R. Rahman	30.7
7	CHANYEOL	27.89
8	Lana Del Rey	21.46

5) Listening Dominance of Top 5 Artists

Analysis shows that the top 5 artists contribute a significant percentage of total listening time, indicating selective listening behavior rather than broad exploration.

top_5_artist_percentage
14.23

6) Artist Diversity Over Time

The number of unique artists listened to per year shows gradual variation, suggesting a balance between discovering new artists and revisiting familiar ones.

	year	unique_artists
1	2022	568
2	2023	842
3	2024	1001
4	2025	953

7) Device Categories Driving Listening Time

Device analysis reveals that one primary device category (such as mobile) dominates listening time, emphasizing on-the-go music consumption.

	device_type	listening_hours
1	Mobile	1081.24
2	PC/Desktop	804.33
3	Other	83.22
4	Smart Sp...	10.81

8) Device Usage Trends Over the Years

Over time, device usage patterns remain consistent, indicating stable listening environments rather than frequent shifts in listening context.

	year	device_type	listening_hours
1	2022	PC/Desktop	151.1
2	2022	Other	76.25
3	2022	Mobile	64.3
4	2022	Smart Sp...	10.81
5	2023	PC/Desktop	280.67
6	2023	Mobile	251.97
7	2023	Other	6.97
8	2024	PC/Desktop	287.31
9	2024	Mobile	266.42
10	2025	Mobile	498.55
11	2025	PC/Desktop	85.24

9) Device Usage by Time of Day

Different devices are preferred at different times of day, suggesting contextual listening behavior based on availability and activity.

	hour	device_type	total_minutes
1	0	PC/Desktop	1.58
2	0	Mobile	189.63
3	1	PC/Desktop	9.27
4	1	Mobile	188.33
5	2	PC/Desktop	107.8
6	2	Mobile	556.57
7	3	PC/Desktop	102.1
8	3	Mobile	2647.99
9	3	Smart Sp...	0.5
10	3	Other	70.72
11	4	PC/Desktop	346.84
12	4	Smart Sp...	21.6
13	4	Mobile	5050.71
14	4	Other	300.67
15	5	PC/Desktop	3130.34
16	5	Smart Sp...	163.81
17	5	Mobile	12279.86
18	5	Other	904.21
19	6	PC/Desktop	6154.37

10) Peak Listening Hours

Listening activity peaks during specific hours of the day, indicating preferred time windows for music consumption tied to daily routines.

	hour	total_minutes
1	6	19902.77
2	7	16558.08
3	5	16478.21
4	14	10698.2
5	15	9665
6	16	7718.01
7	8	6793.3
8	17	6097.94
9	4	5719.82
10	13	3589.01
11	18	3187.14
12	3	2821.32
13	9	2382.78
14	10	1861.59
15	12	1400
16	11	1324.7
17	19	970.24
18	2	664.37
19	20	379.29
20	1	197.6
21	0	191.21
22	21	93.99
23	22	41.27
24	23	40.61

11) Weekday vs Weekend Listening Behavior

Listening hours are higher during weekends, suggesting Spotify is used more extensively during leisure time compared to workdays.

day_type	listening_hours
Weekend	547.36
Weekday	1432.24

12) Most Replayed Tracks

Replay analysis shows that certain tracks are played repeatedly, reflecting strong emotional or habitual attachment to specific songs.

	track	artist	play_count
1	Idea 10	Gibran Alcocer	1192
2	Forever Star(《偷偷藏不住》电视剧插曲)	張洢豪	228
3	LAS ESTRELLAS	SUPER SO...	223
4	無羈 (器樂版)	Lin Hai	211
5	Afternoon Lady II	Park Jin Huyn	187
6	Everytime - Instrumental	CHEN	178
7	Stay With Me - Instrumental	CHANYEOL	175
8	Those Eyes	New West	164
9	Everlast	Will Van De ...	155
10	Scars leave beautiful trace - Instrum...	Car, the gard...	140

13) Tracks with Highest Listening Time

Some tracks accumulate high total listening time despite fewer replays, indicating longer engagement per session rather than frequent repetition.

	track	artist	total_minutes
1	Idea 10	Gibran Alcocer	2420.02
2	LAS ESTRELLAS	SUPER SOUND	784.25
3	Forever Star(《偷...	張洢豪	775.02
4	Those Eyes	New West	607.59
5	Stay With Me - I...	CHANYEOL	561.5
6	Everytime - Instr...	CHEN	559.99
7	Afternoon Lady II	Park Jin Huyn	540.81
8	Cinnamon Girl	Lana Del Rey	531.24
9	無羈(器樂版)	Lin Hai	527.77
10	Go away go awa...	CHANYEOL	522.57

14) Most Common Listening Hour by Device

Each device category exhibits a distinct peak listening hour, reinforcing the idea that listening behavior varies based on context and device availability.

	device_type	hour	total_minutes	rnk
1	Mobile	5	12279.86	1
2	Other	6	1324.85	1
3	PC/Desktop	14	8361.75	1
4	Smart Sp...	6	164.39	1

15) Skip Behavior by Time of Day

Skip rate analysis shows higher skipping during certain parts of the day, suggesting more distracted or passive listening during those periods.

	day_part	total_listens	total_skips	skip_percentage
1	Afternoon	10337	630	6.09
2	Night	2852	164	5.75
3	Evening	3164	181	5.72
4	Morning	19229	951	4.95

6. Dashboard in Power BI

The Power BI dashboard was designed across **three analytical pages**, focusing on overall listening behavior, music preferences, artist discovery trends, time-based listening patterns, and device usage. The dashboard enables interactive exploration of listening habits across time, artists, tracks, and platforms.

- **Page 1 — Overview**

- **Page 2 — Sound Capsule**

➢ Page 3 — Sessions

Primary Insights

The dashboard revealed several key behavioral patterns:

Listening Volume & Engagement

- Total listening time exceeded **1,900+ hours** across the analysis period, indicating strong and consistent engagement with Spotify.
- Average daily listening time was approximately **100+ minutes**, reinforcing Spotify's role as a daily habit rather than occasional usage.

- **Listening Trends Over Time**
  - Year-over-year analysis showed **stable listening growth**, with no sharp drops, suggesting sustained interest rather than short-term spikes.
  - Peak listening volumes were observed in recent years, reflecting increased platform usage over time.
- **Artist Preference & Concentration**
  - The **top 5 artists contributed a significant share of total listening time**, demonstrating strong preference concentration.
  - Despite this concentration, over **1,900+ unique artists** were streamed, indicating balanced exploration alongside repeat listening.
- **Top Tracks & Replay Behavior**

    - Certain tracks recorded **high replay counts**, highlighting emotional attachment or habitual listening.
    - Some tracks accumulated high total listening minutes despite fewer replays, indicating longer session-based engagement rather than frequent repetition.
- **Time-of-Day Listening Patterns**
    - Listening activity peaked during **morning and evening hours**, aligning with daily routines such as commuting, work, and relaxation.
    - Late-night listening showed lower overall volume but higher session depth, suggesting focused listening periods.
- **Weekday vs Weekend Behavior**
    - Weekend listening hours were consistently higher than weekdays, indicating increased leisure-time consumption.
    - Weekday listening remained steady, suggesting background or routine-based music usage during workdays.
- **Seasonal Listening Distribution**
    - **Spring and Winter seasons** accounted for the highest listening minutes, highlighting seasonal variation in music consumption.
    - Summer months showed comparatively lower engagement, possibly due to lifestyle or activity changes.
- **Device Usage Patterns**
    - **Mobile devices dominated listening time**, contributing the largest share of total hours and reinforcing on-the-go consumption.
    - Desktop and smart speaker usage played a secondary role, often tied to specific time windows or listening contexts.
- **Device Trends Over Time**
    - Device usage patterns remained **relatively stable across years**, suggesting consistent listening environments rather than frequent platform switching.
    - Mobile usage increased steadily, while other devices-maintained niche but consistent usage.
- **Discovery Behavior**

    - Monthly new artist discovery showed **noticeable peaks during mid-year months**, indicating higher exploration during certain periods.
    - Discovery activity slowed during other months, suggesting alternating cycles of exploration and familiarity.
- **Engagement & Skip Behavior**
    - Skip rates varied by **day part**, with higher skipping observed during distracted periods such as mornings.
    - Evening and night sessions showed **lower skip percentages**, indicating more intentional and focused listening.

7. Challenges Faced

During the project, several technical challenges were encountered:

- Cleaning semi-structured JSON streaming logs while preserving analytical accuracy.
- Standardizing highly granular platform values into meaningful device categories.
- Handling missing artist metadata returned by the Spotify Web API.
- Engineering time-based features that accurately represented listening behavior.
- Designing an intuitive dashboard capable of presenting multiple behavioral perspectives without overwhelming the user.